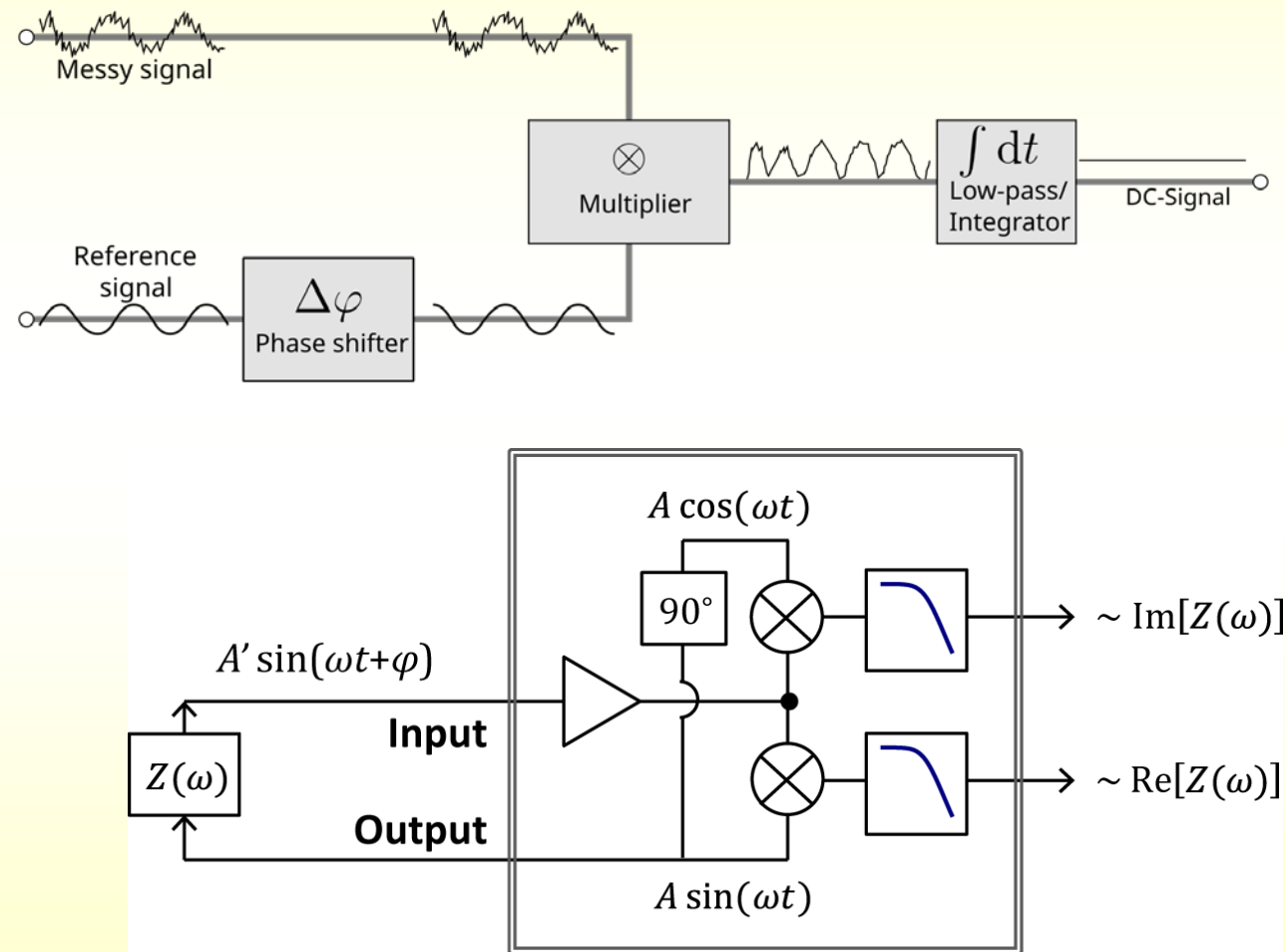


The Lock-in amplifier (for Meas. 11, Measurement Techniques Lab)

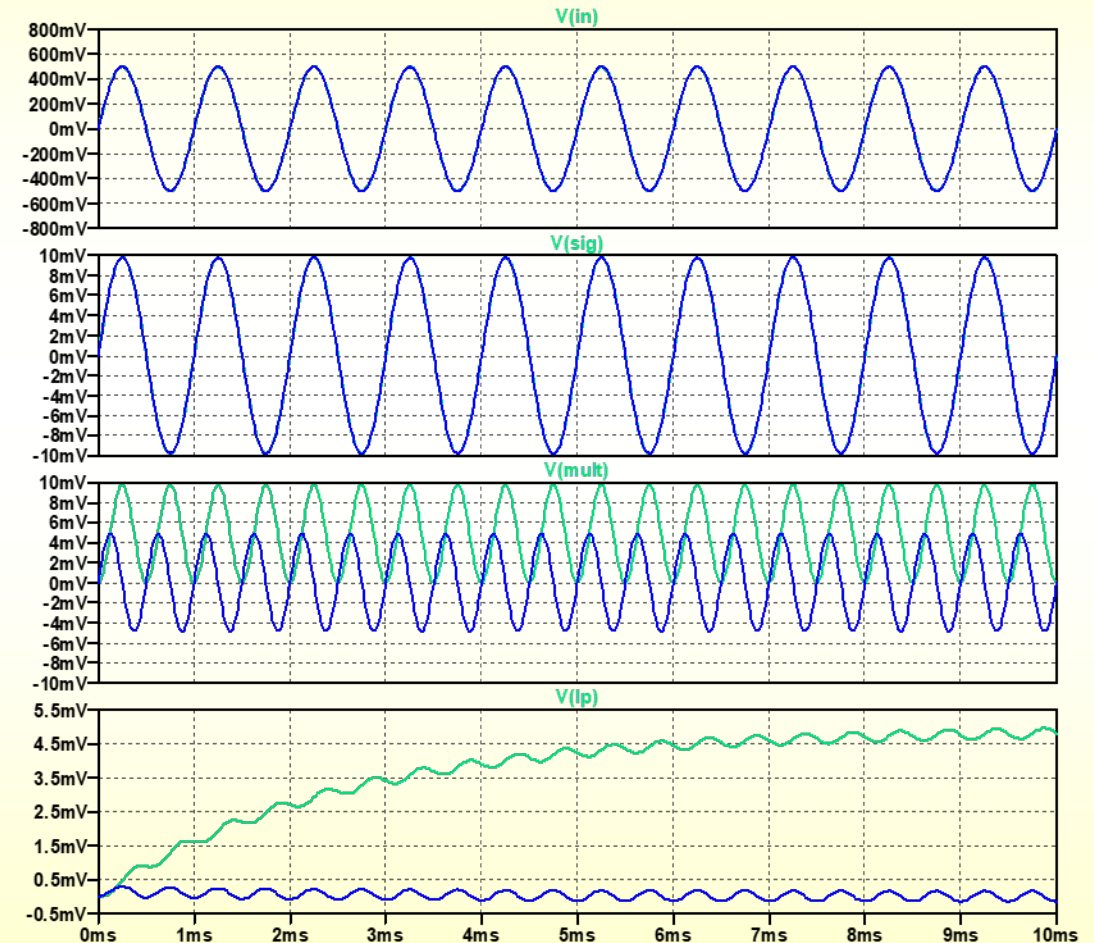
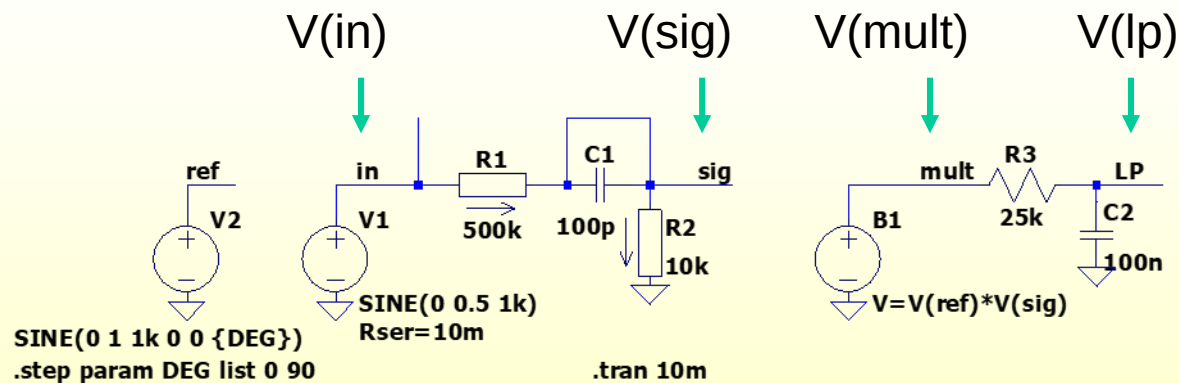




The Lock-in amplifier (for Meas. 11, Measurement Techniques Lab)

Series R circuit measurement

- $f=1$ kHz, reference: $\text{phase}=0^\circ$, $\text{phase}=90^\circ$
- C1 is shorted: only **R1** is measured
- Current is measured as voltage on R2
- B1 **mixer**: multiplies signal and reference
- R3*C2 filters high frequency: lock-in shows $V(\text{lp})$

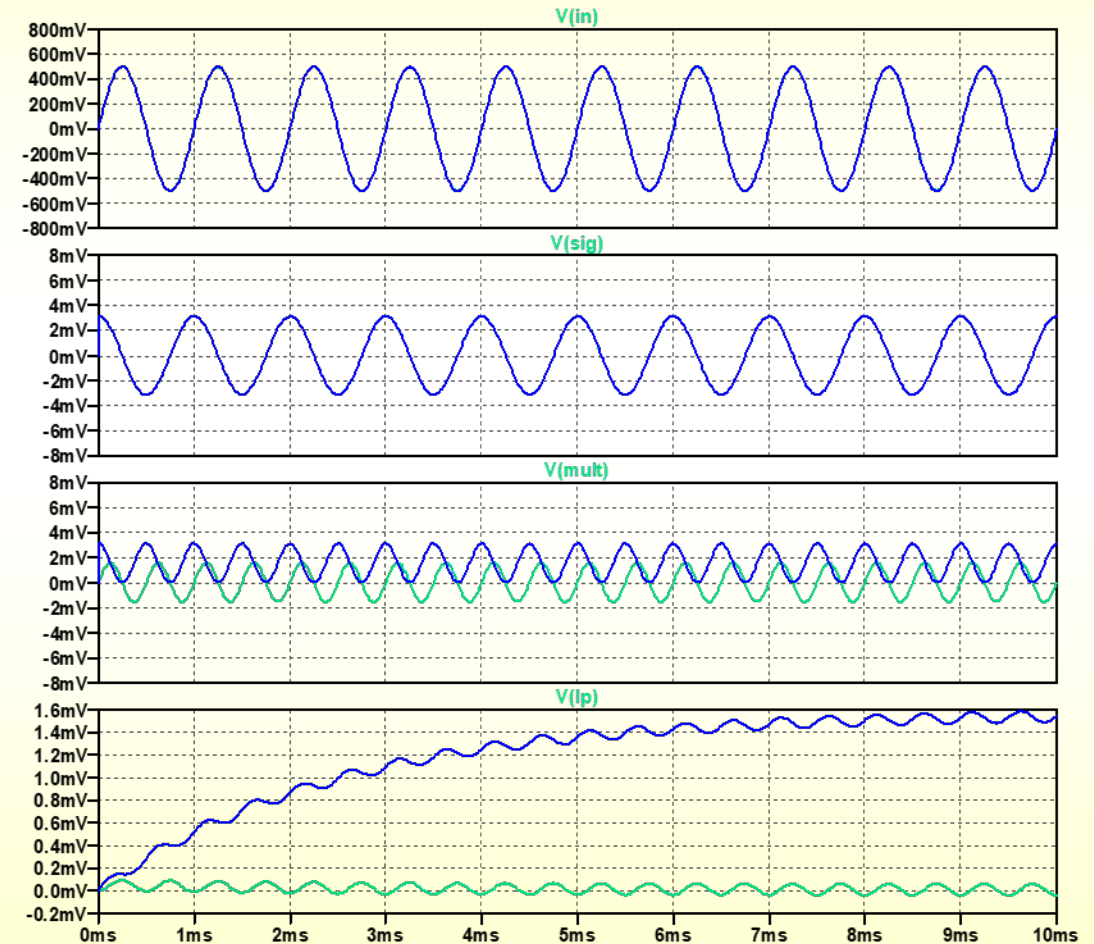
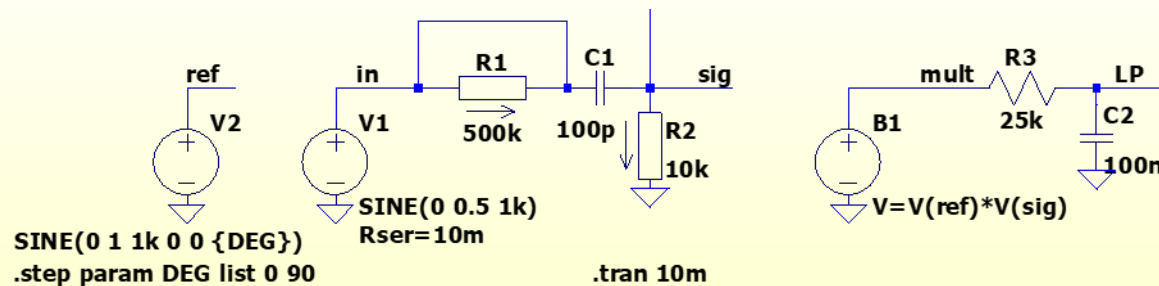




The Lock-in amplifier (for Meas. 11, Measurement Techniques Lab)

Series C circuit measurement

- $f=1$ kHz, reference: $\text{phase}=0^\circ$, $\text{phase}=90^\circ$
- R1 is shorted: only **C1** is measured

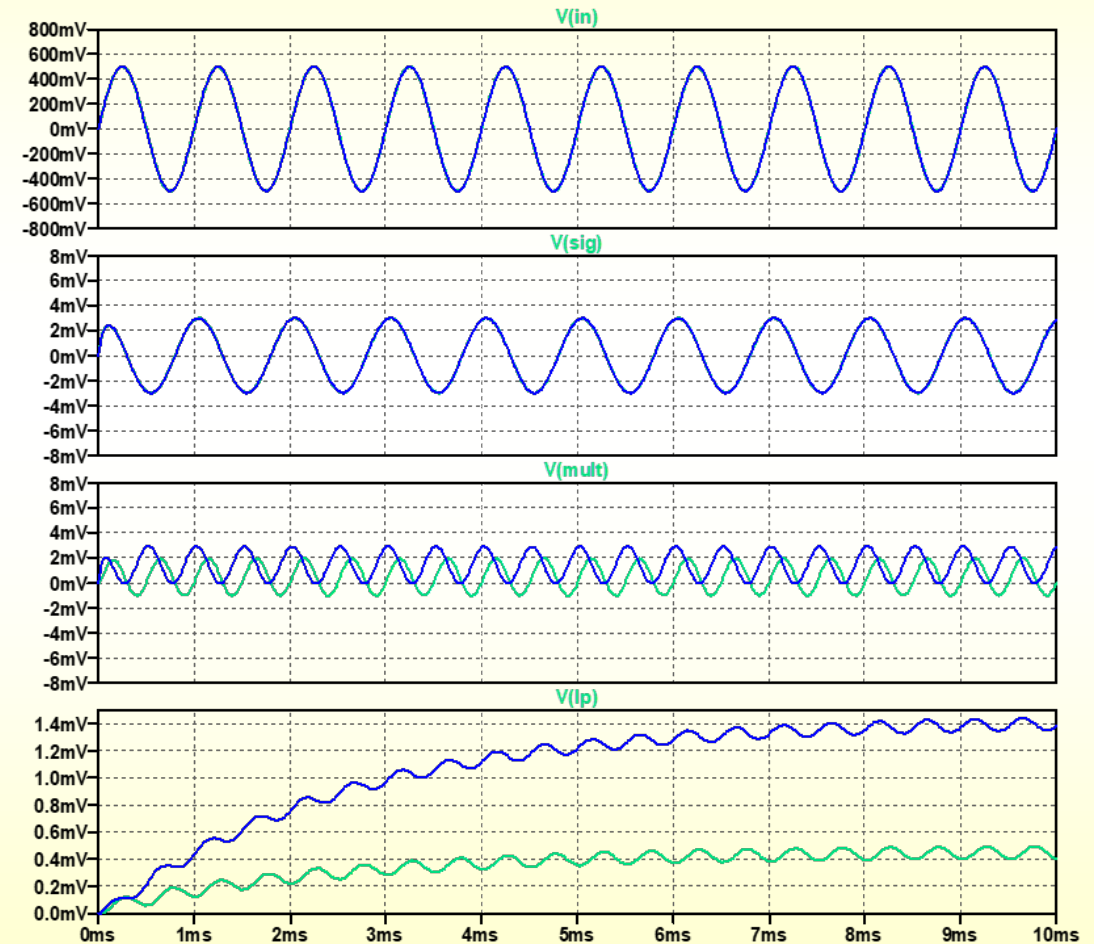
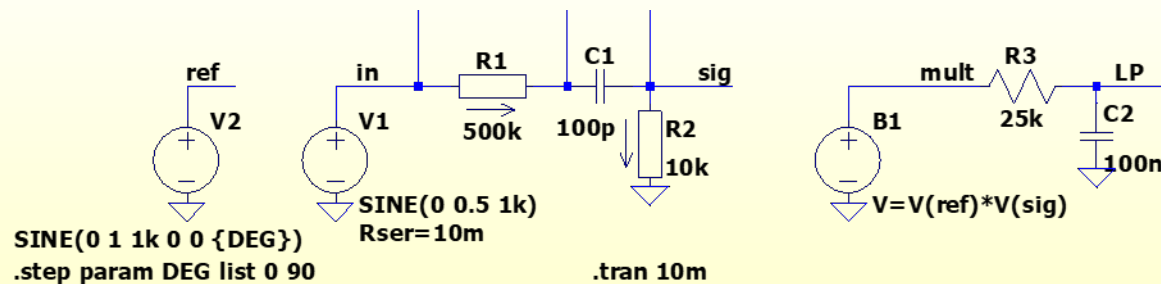




The Lock-in amplifier (for Meas. 11, Measurement Techniques Lab)

Series RC circuit measurement

- $f=1$ kHz, reference: $\text{phase}=0^\circ$, $\text{phase}=90^\circ$
- Series **R1** and **C1** are measured

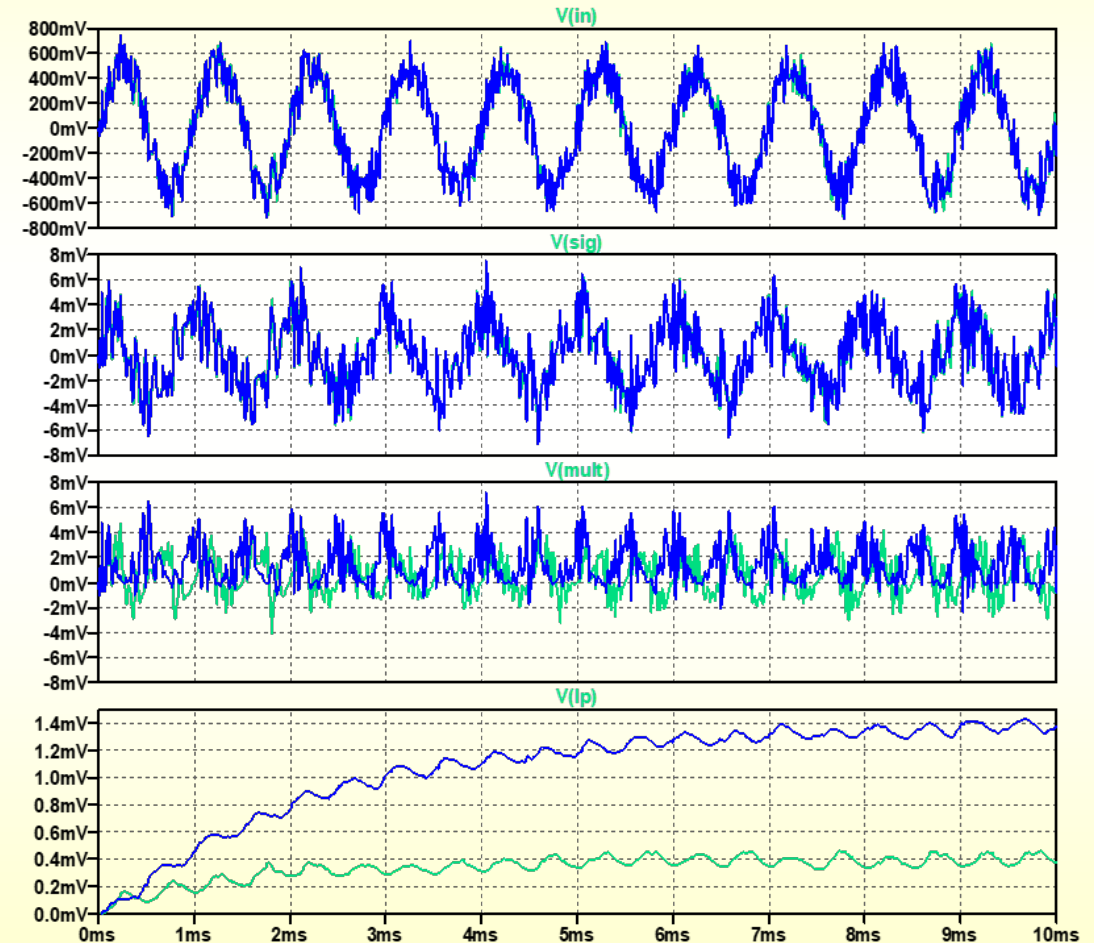
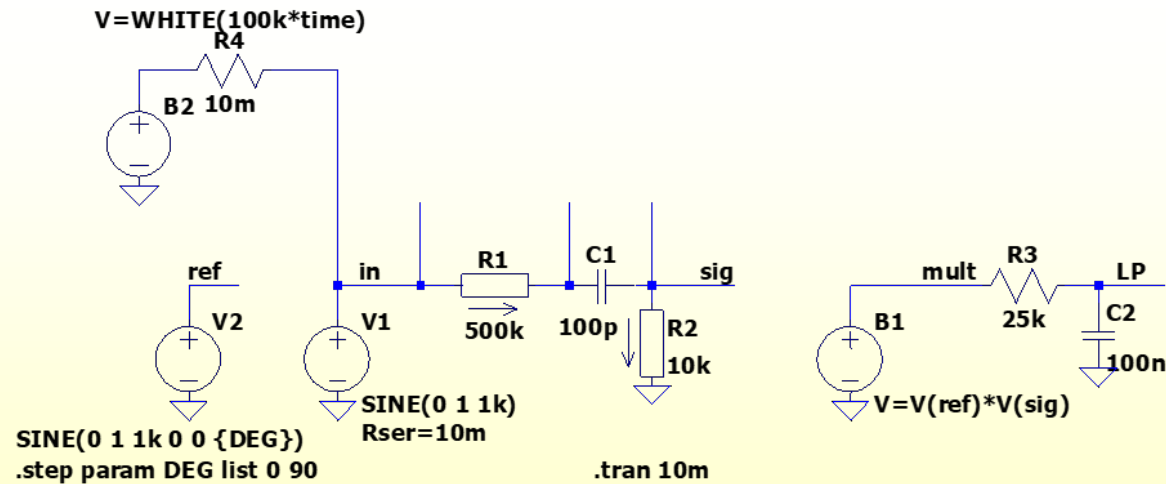




The Lock-in amplifier (for Meas. 11, Measurement Techniques Lab)

Series RC circuit measurement

- $f=1$ kHz, reference: $\text{phase}=0^\circ$, $\text{phase}=90^\circ$
- With white noise generation
- No significant noise on $V(\text{LP})$ output!

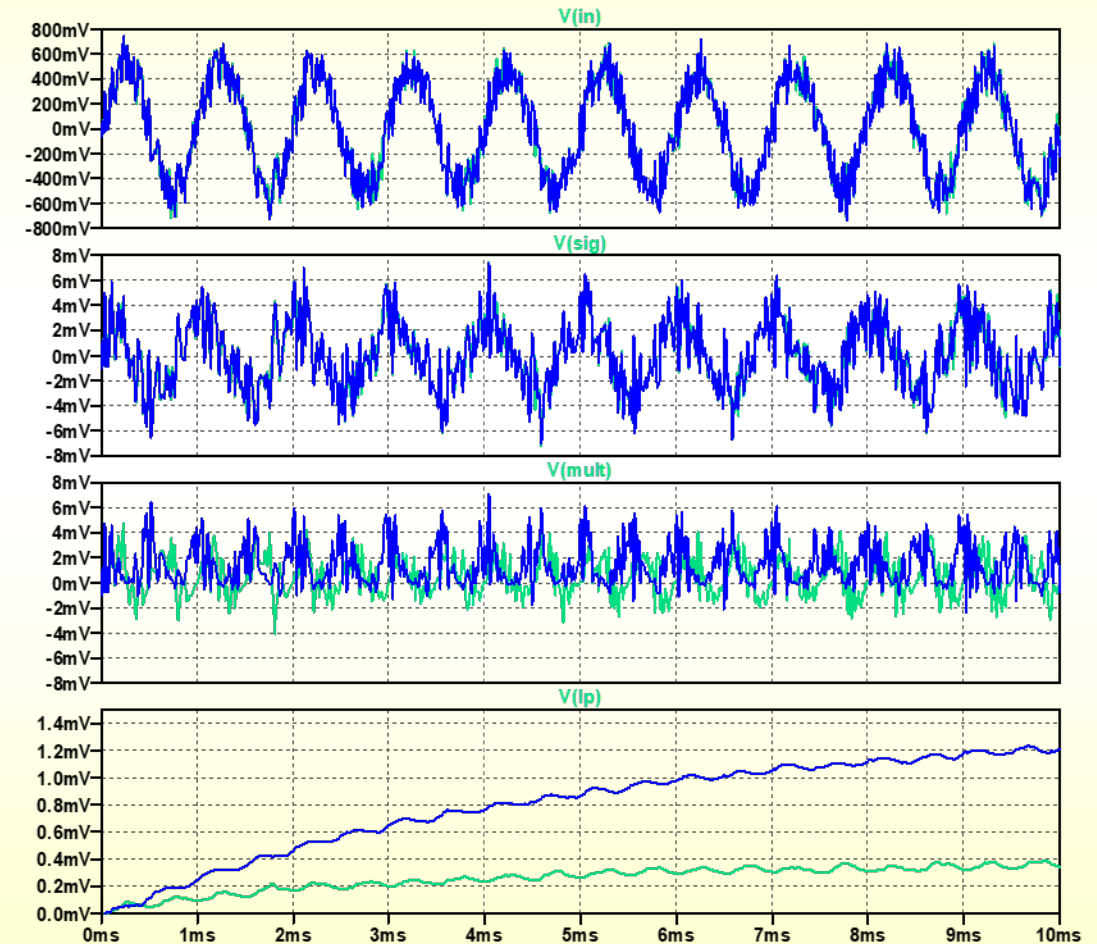
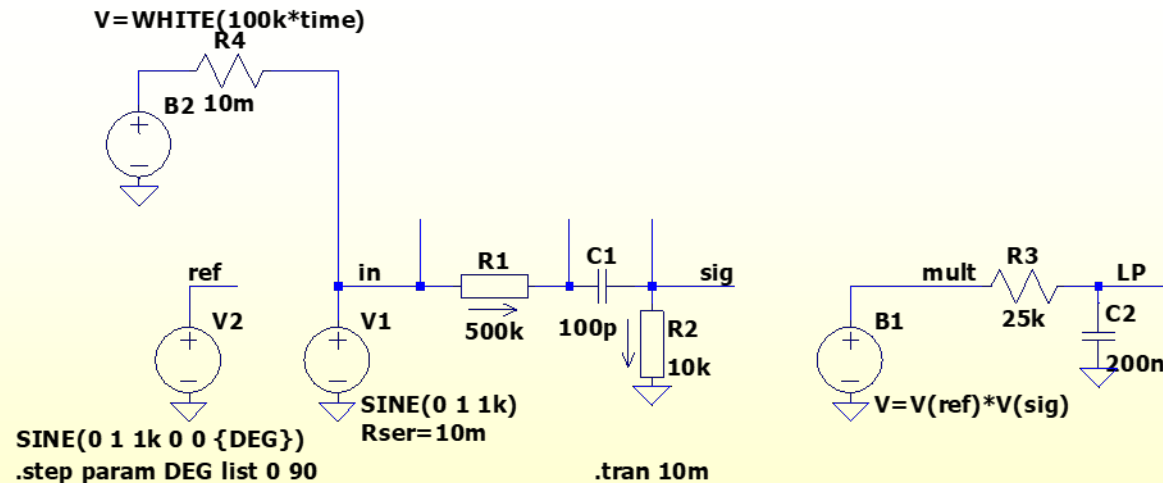




The Lock-in amplifier (for Meas. 11, Measurement Techniques Lab)

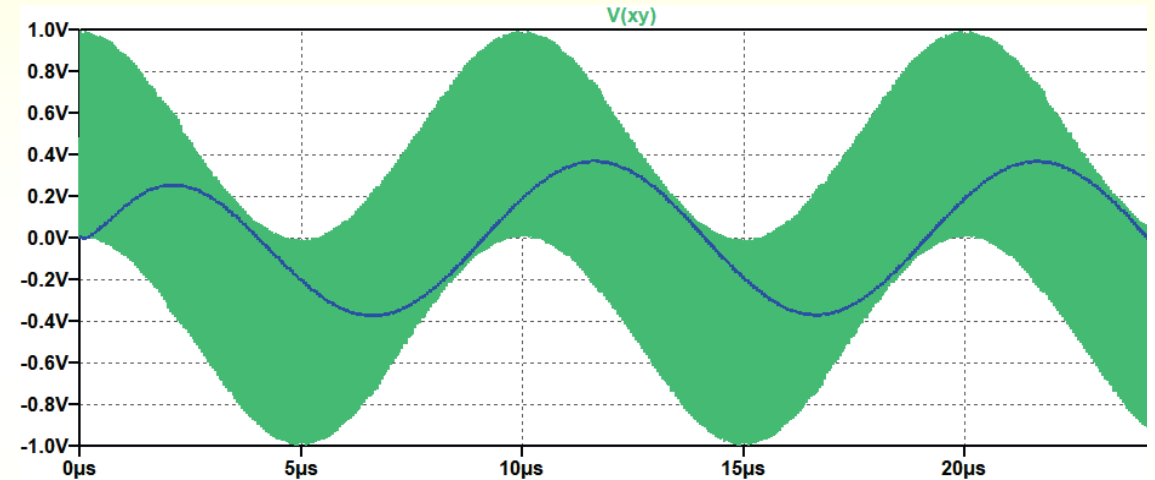
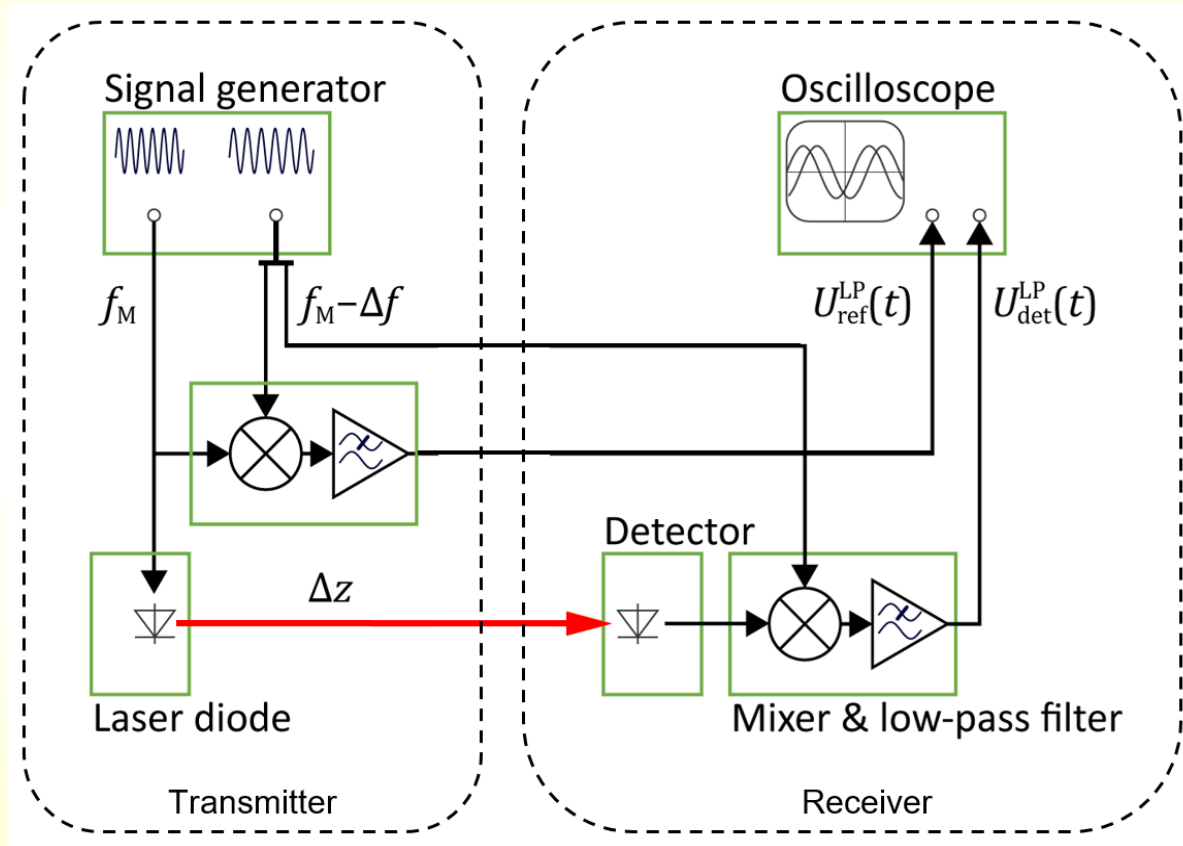
Series RC circuit measurement

- $f=1$ kHz, reference: $\text{phase}=0^\circ$, $\text{phase}=90^\circ$
- If $R3 \cdot C2$ integration time increases:
 - residual high frequency decreases
 - settling time increases





The analog multiplier (for Meas. 10, Measurement Techniques Lab)



- $V(xy)$: multiplier output
- Low-pass filtered output beat signals: $U_{ref}^{LP}(t)$ and $U_{det}^{LP}(t)$
- (Lock-in: mixer multiplies signals of identical frequencies!)